AI-POWERED SMART TRAFFIC ANALYTICS DASHBOARD

Real-Time Computer Vision | Adaptive Signal Control | Predictive Analytics

Category: AI Systems | Smart City Mission | Final Year Project

PROBLEM STATEMENT

Urban traffic congestion affects millions daily — causing fuel waste, pollution & productivity loss. Traditional fixed-timer signals fail to adapt to real-time conditions. Cities like Mumbai & Bengaluru face severe gridlock during peak hours with no intelligent, data-driven traffic control in place.

OUR SOLUTION

An intelligent real-time traffic management system using Computer Vision (YOLO), Machine Learning, and a centralized web dashboard. Works with existing CCTV cameras — zero new hardware required. Built with Python, OpenCV, TensorFlow, React.js and deployed on cloud infrastructure.

KEY FEATURES

Real-Time Vehicle Detection
YOLO-based CV counts vehicles by type & density

Adaptive Signal Control
AI adjusts signal timings based on live traffic data

Emergency Vehicle Priority
Auto-clears paths for ambulances & fire trucks

Predictive Analytics
ML forecasts congestion before it happens

Live Admin Dashboard
Heatmaps, alerts & stats on a single screen

Zero Extra Hardware
Works with existing CCTV infrastructure

TECH STACK: Python | OpenCV | YOLO (Deep Learning) | TensorFlow | React.js | Node.js | Azure Cloud | REST APIs

Database: MongoDB | Visualization: D3.js | CV Model: YOLOv8 | Deployment: Docker + Azure

40%
Reduction in Signal Wait Time

30%
Lower Fuel Consumption

50%
Faster Emergency Response

100%
Existing CCTV Compatible